

Lecture 22: Eigenvalue and Eigenvector (first part)

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[Based on slides for „Decoding the Matrix“]



WEEK

TWELVE



Agenda

Chapter 12

Lecture 22:

Dynamic systems
Eigenvalue and eigenvectors
Diagonalization
Interpretation of basis change
Solving dynamic systems for large values





Lecture 21 - Summary

Previous

Best rank-one approximation

Singular Value Decomposition (SVD)

First Singular
value, Right
Singular vector,
Left Singular
vector

Solving $Ax = b$

Best rank-k approximation

Concept & Definition



Two interest bearing account

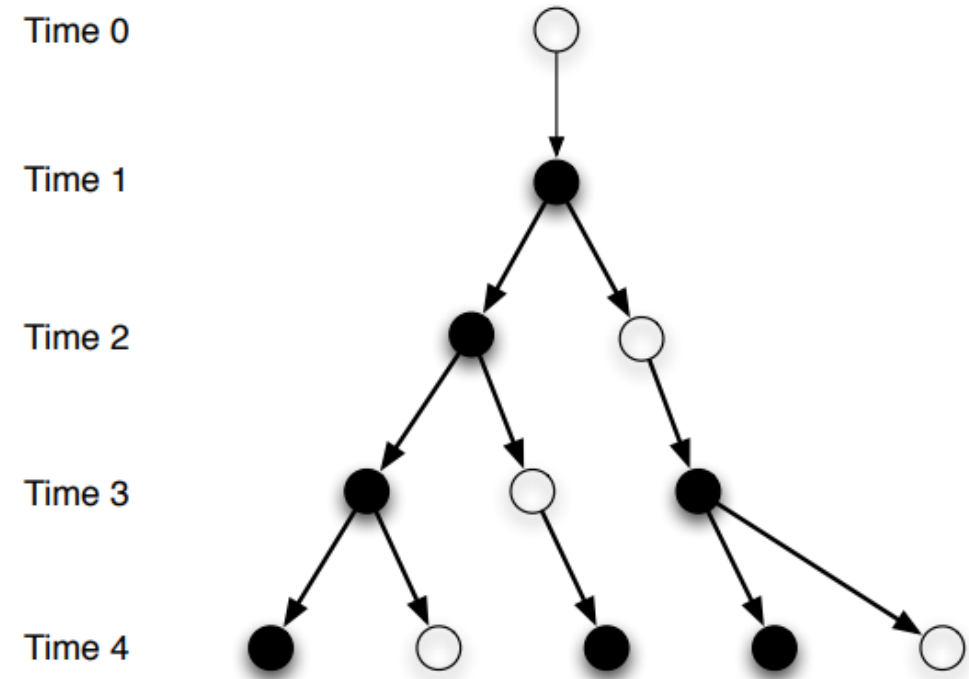
Suppose Account 1 yields 5% interest and Account 2 yields 3% interest.

Represent balances in the two accounts by a 2-vector $\mathbf{x}^{(t)} = \begin{bmatrix} \text{amount in Account 1} \\ \text{amount in Account 2} \end{bmatrix}$.

Rabbit reproduction

To avoid getting into trouble, I'll pretend sex doesn't exist.

- ▶ Each month, each adult rabbit gives birth to one baby.
- ▶ A rabbit takes one month to become an adult.
- ▶ Rabbits never die.



$$\begin{bmatrix} \text{adults at time } t+1 \\ \text{babies at time } t+1 \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_A \begin{bmatrix} \text{adults at time } t \\ \text{babies at time } t \end{bmatrix}$$

Analyzing rabbit reproduction



Fact 12.1.1: Let $S = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} \\ 1 & 1 \end{bmatrix}$. Then $S^{-1}AS$ is the diagonal matrix $\begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix}$

$$\begin{pmatrix} 354224848179261915075 \\ 218922995834555169026 \end{pmatrix}$$



Interpreting using change of basis

Interpretation: To make the analysis easier, we will use a change of basis

Basis consists of the two columns of the matrix S , $\mathbf{v}_1 = \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$

Let $\mathbf{u}^{(t)}$ = coordinate representation of $\mathbf{x}^{(t)}$ in terms of $\mathbf{v}_1$ and $\mathbf{v}_2$.

- ▶ (rep2vec) To go from repres. $\mathbf{u}^{(t)}$ to vector $\mathbf{x}^{(t)}$ itself, we multiply $\mathbf{u}^{(t)}$ by S .
- ▶ (Move forward one month) To go from $\mathbf{x}^{(t)}$ to $\mathbf{x}^{(t+1)}$, we multiply $\mathbf{x}^{(t)}$ by A .
- ▶ (vec2rep) To go to coord. repres., we multiply by S^{-1} .

Eigenvalue and eigenvectors



Definition: If λ is a scalar and $\mathbf{v}$ is a nonzero vector such that $A\mathbf{v} = \lambda\mathbf{v}$, we say that λ is an *eigenvalue* of A , and $\mathbf{v}$ is a corresponding *eigenvector*.

Eigenvectors corresponding to eigen value



Similarity



Definition: Two matrices A and B are *similar* if there is an invertible matrix S such that $S^{-1}AS = B$.

Diagonalizability (1)



Definition: If A is similar to a diagonal matrix, i.e. if there is an invertible matrix S such that $S^{-1}AS = \Lambda$ where Λ is a diagonal matrix, we say A is *diagonalizable*.

Diagonalizability (2)



Diagonalizability Theorem: An $n \times n$ matrix A is diagonalizable iff it has n linearly independent eigenvectors.

Interpretation using change of basis, revisited



$$\begin{bmatrix} \mathbf{u}^{(t)} \end{bmatrix} = \begin{bmatrix} \lambda_1^t & & \\ & \ddots & \\ & & \lambda_n^t \end{bmatrix} \begin{bmatrix} \mathbf{u}^{(0)} \end{bmatrix}$$

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n$$

$$\begin{aligned} A\mathbf{x} &= A(\alpha_1 \mathbf{v}_1) + \cdots + A(\alpha_n \mathbf{v}_n) \\ &= \alpha_1 A\mathbf{v}_1 + \cdots + \alpha_n A\mathbf{v}_n \\ &= \alpha_1 \lambda_1 \mathbf{v}_1 + \cdots + \alpha_n \lambda_n \mathbf{v}_n \end{aligned}$$

$$A^t \mathbf{x} = \alpha_1 \lambda_1^t \mathbf{v}_1 + \cdots + \alpha_n \lambda_n^t \mathbf{v}_n$$

Interpretation using change of basis, re-revisited



$$A^t \mathbf{x} = \alpha_1 \lambda_1^t \mathbf{v}_1 + \cdots + \alpha_n \lambda_n^t \mathbf{v}_n$$



Rabbit reproduction and death

A disease enters the rabbit population. In each month,

- ▶ A δ fraction of the adult population catches it,
- ▶ an ϵ fraction of the juvenile population catches it, and
- ▶ an η fraction of the sick population die, and the rest recover.

Sick rabbits don't produce babies.

Equations

$$\text{well adults}' = (1 - \delta) \text{ well adults} + (1 - \epsilon) \text{ juveniles} + (1 - \eta) \text{ sick}$$

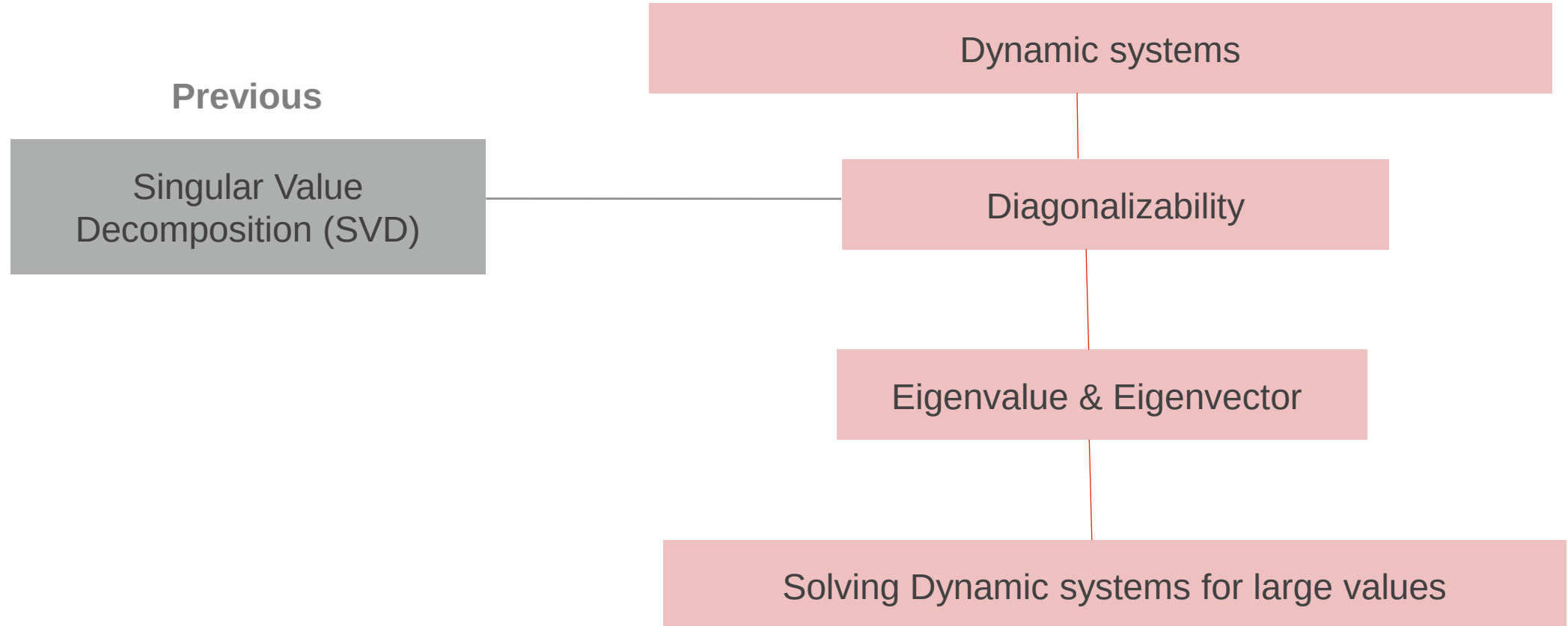
$$\text{juveniles}' = \text{well adults}$$

$$\text{sick}' = \delta \text{ well adults} + \epsilon \text{ juveniles} + 0$$



Lecture 22 - Recap

Concept & Definition





Thank You For Your Attention!





Theoretical exercise

Problem: Let A be a diagonalizable matrix with λ a non zero eigenvalue for A , using λ express the eigenvalue of A^{-1}

Hint: $Ax = \lambda x$, and left multiply by A^{-1}



Applied exercise

Problem: Let be $A = [a_1 \mid a_2 \mid a_3]$ a matrix with $a_1 = (2,0,0)$, $a_2 = (-1,1,0)$ and $a_3 = (9,7,3)$, prove that 2,1 and 3 are the eigenvalues of A.